A Look at GNU Unified Parallel C

This article guides readers through the installation and usage of GNU Unified Parallel C

GNU Unified Parallel C is an extension to the GNU C Compiler (GCC), which supports the execution of Unified Parallel C (UPC) programs. UPC uses the Partitioned Global Address Space (PGAS) model for its implementation. The current version of UPC is 1.2, and a 1.3 draft specification is available. GNU UPC is released under the GPL license, while the UPC specification is released under the new BSD license. To install it on Fedora, you need to first install the *gupc* repository:

```
$ sudo yum install http://www.gccupc.org/pub/pkg/rpms/gupc-
fedora-18-1.noarch.rpm
```

You can then install the *gupc* RPM using the following command:

```
$ sudo yum install gupc-gcc-upc
```

The installation directory is */usr/local/gupc*. You will also require the *numactl* (library for tuning Non-Uniform Memory Access machines) development packages:

```
$ sudo yum install numactl-devel numactl-libs
```

To add the installation directory to your environment, install the environment modules package:

```
$ sudo yum install environment-modules
```

You can then load the *gupc* module with:

```
# module load gupc-x86_64
```

Consider the following simple 'hello world' example:

```
#include <stdio.h>
int main()
{
printf ("Hello World\n");
return 0;
}
```

You can compile it using:

```
# gupc hello.c -o hello
Then run it with:
```

```
# ./hello -fupc-threads-5

Hello World
Hello World
Hello World
Hello World
Hello World
```

The argument *-fupc-threads-N* specifies the number of threads to be run. The program can also be executed using:

```
# ./hello -n 5
```

The *gupc* compiler provides a number of compile and run-time options. The '-v' option produces a verbose output of the compilation steps. It also gives information on GNU UPC. An example of such an output is shown below:

```
# gupc hello.c -o hello -v

Driving: gupc -x upc hello.c -o hello -v -fupc-link
```

Using built-in specs.

```
COLLECT_GCC=gupc
COLLECT_LTO_WRAPPER=/usr/local/gupc/libexec/gcc/x86_64-
redhat-linux/4.8.0/lto-wrapper
Target: x86_64-redhat-linux

Configured with: ...
Thread model: posix
gcc version 4.8.0 20130311 (GNU UPC 4.8.0-3) (GCC)
COLLECT_GCC_OPTIONS='-o' 'hello' '-v' '-fupc-link'
'-mtune=generic' '-march=x86-64'
...

GNU UPC (GCC) version 4.8.0 20130311 (GNU UPC 4.8.0-3)
(x86_64-redhat-linux)
compiled by GNU C version 4.8.0 20130311 (GNU UPC 4.8.0-3),
GMP version 5.0.5, MPFR version 3.1.1, MPC version 0.9
GGC heuristics: --param ggc-min-expand=100 --param ggc-min-
heapsize=131072
...
#include "..." search starts here:
#include <...> search starts here:
/usr/local/gupc/lib/gcc/x86_64-redhat-linux/4.8.0/include
```